

Task: Descriptive Statistics for Beginner Data Analysis & Data Scientists

Descriptive Statistics for Data Analysis & Machine Learning

1) Measures of Central Tendency (Finding the center of Data)

1. Mean

- The average value of a set of data, calculated by dividing the sum of all values by the total number of values.
- It provides a central numerical summary that takes into account the exact value of every observation in the dataset.

$$\text{mean} = \frac{\text{Sum of all values}}{\text{Number of values}} = \frac{\sum x_i}{n}$$

Ex! Describe the performance of students in a Python module using their test scores: 38, 40, 42, 44, 46, 50, 54, 56, 58.

$$\frac{\sum x_i}{n} = \frac{38+40+42+44+46+50+54+56+58}{9} \Rightarrow \frac{428}{9} = 47.5$$

- Daily Life example: calculating the average monthly electricity bill to plan a household budget.
- Practical use case & Limitation:
 - Use case: Quality control checks to evaluate the average dimension or weight of manufactured parts.
 - Limitation: Easily affected by extreme high or low values

2. Median

- The middle value in a dataset when the values are arranged in ascending or descending order.
- It gives a reliable central measure that is not affected by extreme values.

- Median = ~~average~~ $\frac{n+1}{2}$ $\frac{n}{2}$ & $\frac{n}{2} + 1$

- Problem: 42, 44, 46, 48, 50, 52, 54, 56, 58

$$\frac{9+1}{2} = 5^{\text{th}} \text{ Position}$$

median $\Rightarrow$ position 5 $\Rightarrow$ 50

- Daily Life: Reporting median house prices in a city to avoid market distortion from luxury mansions.

- Use Case & Limitation:

• Use Case: Assessing mid-level income distributions or software latency metrics.

• Limitation: Disregards the exact numerical magnitude of values other than their relative ordering.

3. Mode

- The value that appears most frequently in a dataset.

- It identifies the most popular or recurring score/category in a dataset.

- Mode = value with maximum frequency

- Problem: 40, 45, 50, 50, 50, 55, 60

40 $\rightarrow$ 1

45 $\rightarrow$ 1

50 $\rightarrow$ 3

55 $\rightarrow$ 1

60 $\rightarrow$ 1

$\rightarrow$ value with highest frequency is 50

$\rightarrow$ Mode 50.

- Daily Life Example: An e-commerce clothing store checking the most ordered shirt size to manage inventory.

- Use Case & Limitation:

• Use Case: stock inventory planning based on customer purchase frequency.

• Limitation: Datasets may have no mode if all values are unique, or multiple modes.

a) what it means to find the center of data

A single value that represents the entire dataset and shows where most values are concentrated.

b) Difference between mean and median

Mean

Median

- Average of all values

- Middle value of ordered data

- $\frac{\sum x}{n}$

- middle observation position

- Affected by outliers

- Not affected by outliers

c) when businesses prefer median over mean, especially for skewed data

Businesses prefer the median when the data is skewed or contains outliers, because extreme values can greatly affect the mean while the median remains stable.

② Measures of Dispersion (Understanding Data spread)

a) Range

- The difference between the highest and lowest values in a dataset.

- Provides a fast, simple boundary check for data spread.

- Range = Max value - Min value

- Problem: 10, 15, 20, 25, 30

Max value = 30, min value = 10

$$\text{range} = 30 - 10 = 20$$

- Daily life example: Reporting daily temperature highs and lows.

- Use case: simple boundary checks for operational thresholds.

- Limitation: Heavily distorted by a single extreme minimum or maximum outlier.

b) Variance

- measures the average squared deviation of each value from the mean.
- Quantifies how far individual data points scatter away from the average

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n}$$

- Problem: 2, 4, 6, 8, 10

- ^{mean} $\mu = \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$

deviation $(x - \mu)$ & squared deviation $(x - \mu)^2$

$$(2-6) = -4 \rightarrow (-4)^2 = 16$$

$$(4-6) = -2 \rightarrow (-2)^2 = 4$$

$$(6-6) = 0 \rightarrow (0)^2 = 0$$

$$(8-6) = 2 \rightarrow (2)^2 = 4$$

$$(10-6) = 4 \rightarrow (4)^2 = 16$$

- sum of squared deviations:

$$\sum (x - \mu)^2 = 16 + 4 + 0 + 4 + 16 = 40$$

- $\text{variance} = \frac{40}{5} = 8$

- Daily life example: Investment volatility evaluation in financial portfolios.

- use case: Foundation for advanced statistical algorithms and machine learning models.

- Limitation: measured in squared units, making direct interpretation unintuitive.

c) Standard Deviation

- measures the average spread of data from the mean in original units.
- restores dispersion measurements back to the original unit scale of the data.

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{n}}$$

Problem:
2, 4, 6, 8, 10

$\mu = 6$, sum of squared deviations $\sum (x - \mu)^2 = 40$

$$\text{variance} = \frac{40}{5} = 8$$

$$\text{standard deviation} = \sqrt{8} \approx 2.83$$

- Daily life example: Quality control in manufacturing to maintain consistent product sizes.
- Use Case: Evaluating risk and return consistency in financial investments.
- Limitation: Influenced by outliers since its calculation relies on the mean.

d) Quartiles & Interquartile Range (IQR)

- Quartiles: Cut-off points dividing ordered data into four equal parts:

Q_1 : 25% of data lies below it

Q_2 : 50% of data lies below it (Median).

Q_3 : 75% of data lies below it.

- IQR: measures the spread of the middle 50% of the data.

→ ~~IQR: measures the spread of the middle 50% of the data.~~
IQR isolates the central majority of the dataset while ignoring extreme tail outliers.

$$IQR = Q_3 - Q_1$$

Problem: Find Q_1 , Q_2 , Q_3 and IQR for dataset:

10, 20, 30, 40, 50, 60, 70, 80.

1. Compute Q_2 (Median of full dataset):

$$\text{Middle pair} = (40, 50) \rightarrow Q_2 = \frac{40+50}{2} = 45$$

2. Compute Q_1 (Median of lower half: 10, 20, 30, 40):

$$Q_1 = \frac{20+30}{2} = 25$$

3. Compute Q_3 (Median of upper half: 50, 60, 70, 80):

$$Q_3 = \frac{60+70}{2} = 65$$

4. Compute IQR:

$$IQR = Q_3 - Q_1 = 65 - 25 = 40$$

- Daily life example: Income distribution analyses to examine typical earnings while excluding extreme wealth outliers.
- Use case: Construction of box plots and automated outlier filtering.
- Limitation: Completely disregards information outside the 25th and 75th percentiles.

a) why knowing only the average is not enough

The average gives only the center of the data but does not show how spread out the values are. Two datasets can have the same average but very different variability.

b) How dispersion measures help understand consistency, variability, and risk.

- Consistency: Lower spread means more consistent data.
- Variability: Higher spread means greater variation.
- Risk: Higher variability often indicates higher risk in business and finance.

c) How IQR is used to detect outliers

$$\text{Lower Limit} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Limit} = Q_3 + 1.5 \times IQR$$

3) Measures of shape (understanding Data distribution)

1. skewness

skewness measures the symmetry of a data distribution.

i) Left-skewed Distribution

- The tail extends to the left, and most values are concentrated on the right.
- $\text{mean} < \text{median} < \text{mode}$.
- Example: 10, 80, 85, 90, 90, 95, 100 (low extreme value 10 pulls mean to the left).
- Age at retirement, where most people retire late, but a few retire early.

ii) Right-skewed Distribution

- The tail extends to the right, and most values are concentrated on the left.
- $\text{mean} > \text{median} > \text{mode}$
- Example: 10, 12, 12, 15, 20, 150 (high outlier 150 pulls mean to the right).
- Household income distribution, where a few top earners create a long right tail.

iii) Symmetric Distribution

- A distribution where both sides are mirror images of each other.

- $\text{mean} = \text{median} = \text{mode}$

- Problem: 10, 20, 30, 40, 50

$$\text{mean} = \frac{10 + 20 + 30 + 40 + 50}{5} = 30$$

$$\text{median} = 30$$

$$\text{mode} = \text{None}$$

This dataset is symmetric because the values are evenly distributed around the center (30).

2) Kurtosis

Kurtosis measures the peakedness and tail heaviness of a distribution.

i) Normal Distribution (mesokurtic)

- A normal distribution has a moderate peak and moderate tails.

- Example: 60, 65, 70, 75, 80, 75, 70, 65, 60

Scores rise smoothly to a central peak at 80 and fall off symmetrically with standard tail thickness.

ii) Heavy-Tailed Distribution (Leptokurtic)

- A heavy-tailed distribution has a higher peak and heavier tails than a normal distribution, meaning it contains more extreme values.

- Example: 10, 11, 12, 13, 14, 15, 80

The value 80 is an outlier, indicating a heavy-tailed distribution.

iii) Light-Tailed Distribution (Platykurtic)

- A light-tailed distribution has a flatter peak and lighter tails than a normal distribution, meaning extreme values are less likely.

- Example: 48, 49, 50, 50, 50, 51, 52

The values are close to the center with no extreme values, indicating light-tailed distribution.